

THREAT INTELLIGENCE CAPSTONE EXERCISE AND SECURITY RISK ASSESSMENT

*Analysis of the Cyber Threat Landscape Targeting the
Anambra State Government, Nigeria*



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INTRODUCTION

The Core Definition

Cyber Threat Intelligence (**CTI**) is evidence-based knowledge, including context, mechanisms, indicators, implications, and actionable advice, about an existing or emerging menace or hazard to assets. Its purpose is to inform organizational decisions regarding the response to that menace or hazard.

In simpler terms, Cyber Threat Intelligence is processed and analyzed information that helps organizations understand:

Who is attacking them (***The Threat Actor***).

Why they are being targeted (***The Motivation***).

How they will be attacked (***The Tactics, Techniques, and Procedures - TTPs***).

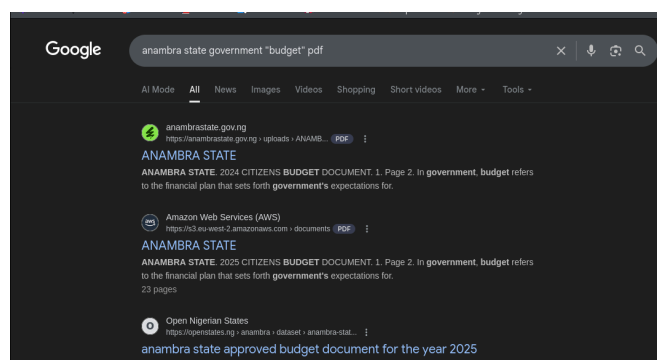
What to do about it (***The Actionable Advice***).

Executive Summary

1. Summary of Findings

This capstone report details the results of a passive Open-Source Intelligence (**OSINT**) gathering exercise and a subsequent Security Risk Assessment targeting the Anambra State Government (ANSG), Nigeria. The analysis was conducted against a backdrop of the State's aggressive e-governance push (e.g., the 'Everything Technology and Technology Everywhere' mantra, rollout of free Wi-Fi, and digitalization of MDAs), which significantly expands its digital attack surface.

Key findings indicate that the ANSG faces Critical Risks primarily stemming from three areas:



Please Note: This report is a simulated academic exercise conducted solely for educational purposes using publicly available Open-Source Intelligence (OSINT).

- **1 Background and Context**

Cyber threats represent a paramount and evolving risk to modern governance, capable of crippling critical services, eroding public trust, and causing significant financial damage. In Nigeria, the cybersecurity threat landscape is characterized by a high volume of sophisticated attacks, with Ransomware, Botnets, and Business Email Compromise (BEC) being particularly prevalent, especially against government institutions.

The Anambra State Government (ANSG) has publicly embarked on an ambitious digital transformation journey, positioning itself as a leader in e-governance within Nigeria. This push includes the digitalization of internal processes, the deployment of public digital services (e.g., e-identity cards, online tax payments), and extensive public Wi-Fi initiatives. While these developments enhance efficiency and transparency, they simultaneously increase the State's exposure to cyber risks.

- **2 Purpose and Scope of the Report**

The purpose of this report is to conduct a simulated, ethical, and passive threat intelligence exercise to identify and analyze the cyber risks confronting the Anambra State Government.

The scope is strictly limited to: Passive Reconnaissance (OSINT): Gathering information solely from publicly available sources (e.g., official websites, public archives, news reports, regulatory filings, and public domain registries). NO active scanning, probing, or unauthorized access was performed.

Threat Landscape Analysis: Identifying the most relevant threat actors and their Tactics, Techniques, and Procedures (TTPs) targeting the Nigerian public sector.

Risk Assessment: Mapping identified public-facing assets and associated vulnerabilities to determine their potential risk and impact on government operations

- **3 Report Structure**

The remainder of this report is structured as follows:

Part One: Details the CTI sources and OSINT tools used for information gathering.

Part Two: Presents the grouped findings of the passive reconnaissance and the comprehensive Security Risk Assessment, including suggested mitigation controls.

Conclusion: Summarizes the overall risk posture and reiterates the strategic recommendations for the ANSG.

MAJOR SOURCES OF CYBER THREAT INTELLIGENCE (CTI)

CTI Source	Type/Relevance	Why it is a Major Source
1. National/Regional CERTs & CISA	Government Alerts/Advisories	Provides information on threats specifically targeting Nigerian/African government entities and general threats (e.g., ransomware) highlighted by global bodies like CISA.
2. Public Vulnerability Databases (NVD, CVE)	Technical Intelligence	To check if any technology publicly known to be used by the state (e.g., specific web server software) has recent, high-severity vulnerabilities.
3. Reputable Cybersecurity News/Blogs (e.g., Nigerian/African Tech News, Foreign Threat Reports)	Strategic/Operational Intelligence	Provides news on recent cyberattacks/breaches affecting Nigerian government bodies or large organizations, which indicates the local threat landscape.
4. Open-Source Threat Feeds (e.g., URLhaus, Abuse.ch)	Tactical Intelligence (IoCs)	To check for Indicators of Compromise (IoCs) related to known Anambra State domains/IPs that may have been reported as malicious (passively).
5. Public Documents/Reports from NITDA Strategic/Tactical Intelligence	The National Information Technology Development Agency	(NITDA) in Nigeria publishes policies and sometimes advisories that outline common attack methods and required controls for government MDAs.
6. The Dark Web/Public Paste Sites Monitoring (using aggregated reports)	Operational Intelligence Reviewing third-party reports (like those from commercial vendors or threat research firms) that summarize leaked credentials or discussions relevant to Nigerian government sectors.	Note: You are NOT accessing the dark web directly, only relying on public reports about it.

THREAT INTELLIGENCE TOOLS USED FOR INFORMATION GATHERING

Specific Tool Example	Tool category	How its used for Anambra State
theHarvester	Passive acquisition of Emails, host & subdomains from search engines	Search engines, Public databases, Employee Email Addresses, Internet Protocols, Subdomains
Google Dorking (e.g site: anambrastate.gov.ng : filetype:pdf confidential)	Advance Search Engine	To find exposed documents, old press releases, staff directories and login pages.
Shodan	Network Information Search	Searching for public services/IPs associated with the known anambrastate.gov.ng domain range to identify accessible services and banners.
Have I Been Pwned (HIBP) Domain Search	Email/Data Breach Checker	Checking for the primary government domain (@anambrastate.ng) to see if any government email addresses has been disclosed in known data breaches
Virus Total	For suspicious file analysing	IPs and URLs malware detector and other breaches,

PROFESSIONAL ORGANISATION/ADVISORY GROUP TO JOIN AND LEARN/EXCHANGE THREAT INTELLIGENCE NEWS

These groups should be relevant for a Nigerian state government to join for CTI sharing.

1. **Cyber Security Experts Association of Nigeria (CSEAN):** A key professional body that promotes cybersecurity best practices and intelligence sharing among Nigerian professionals.
2. **Information Security Society of Africa - Nigeria (ISSAN):** Dedicated to the protection of Nigeria's cyberspace and stands as a major forum for security engagement.
3. **National Information Technology Development Agency (NITDA):** The government body that sets standards and issues advisories for all government Ministries, Departments, and Agencies (MDAs).
4. **A Sector-Specific ISAC (Information Sharing and Analysis Center)** if one exists for Government/Public Sector in Nigeria/ECOWAS: While a dedicated Nigerian Government ISAC might be complex, joining a regional equivalent or working with a major financial/telecom ISAC (as many state functions touch finance) would be beneficial.
5. **FIRST (Forum of Incident Response and Security Teams):** A global leader in incident response and threat sharing; the Anambra State CERT (if they have one) or ICT Agency could join for high-level technical CTI.

SECURITY RISK ASSESSMENT

From the passive use of the threat intelligence tools, Below is a table displaying my findings

Discovery	Source	Vulnerability - weaknesses	Threats - potential harm	Potential Risk - impact of the harm	Risk Mitigation	Risk Severity
E-mails - multiple publicly visible staff emails (.gov.ng)	theHarvester Hunter.io haveibeenpwned.com	Exposed emails can be breached	This could increase the risk of phishing emails.	The threats could produce the risk of account compromise and malware infection, through ransomware.	Implementing MFA, using complex passwords, and email scanners would mitigate the risks mentioned.	High
IP Addresses	theHarvester Shodan.io	IP Addresses revealing open ports that expose databases and web servers .	Attackers can easily gain access and inject malware, or launch DDOS attacks	Data theft, disruption of services, financial loss and possibly death are all potential risks to the threats.	Close idle ports, using VPN tunnels, configuring firewall rules	High
Domain Certificate s/ Subdomain - discovered from public certificate logs.	crt.sh	Multiple wildcard domains with expired certificates were found.	Bruteforced password & DDoS attacks could be launched on the domains.	Subdomains could be retrieved and unauthorized access to the network can be gained	Certificate Monitoring and decommissioning unused domains.	Medium

The "OSINT findings showed a moderate exposure of confidential data that is publicly accessible. Risk mitigation strategies are to be implemented immediately."

The image below shows the results of an Open Source Intelligence (OSINT) investigation, performed using theHarvester tool.

```
File Actions Edit View Help
ansippa@anambra
greg.ezeilo@anambra
ict@anambra
info@anambra
julius.chukwue
sid@anambra

[*] No people found.
[*] Hosts found: 176

*.anambra
*.anambra
*.ascep.anambra 42.244
*.telemed.anambra 64.37.80
```

The resulting IPs address list was enumerated using theHarvester tool.

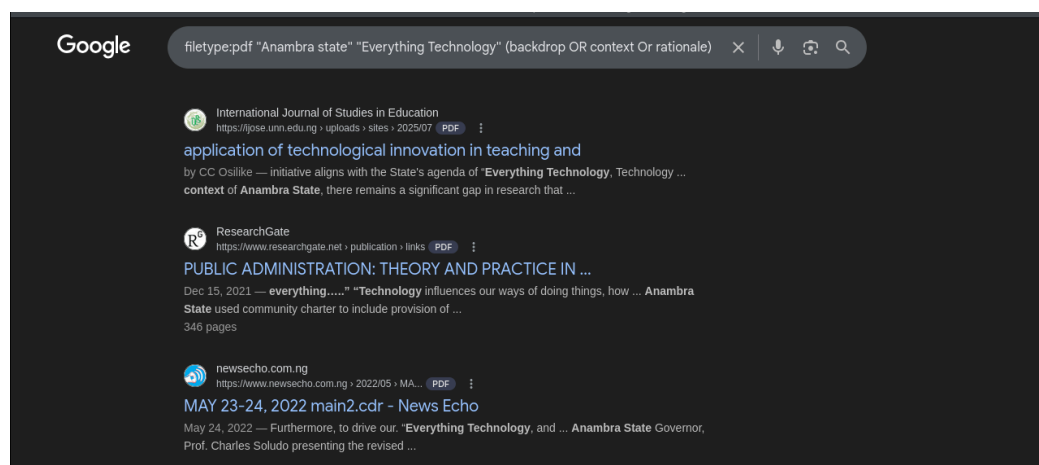
```
[*] IPs found: 10

102.102.102.7.80
102.102.102.3.125
102.102.102.3.81
102.102.102.3.84
102.102.102.3.96
102.102.102.207
141.141.141.4.198
172.172.172.3.111
2606:2606:2606:3031::ac43:d56f
2606:2606:2606::6810:f44e
```

The statement above confirms that the information gathered is specifically the list of email addresses, subdomains, and IP addresses. This is considered a moderate security exposure.

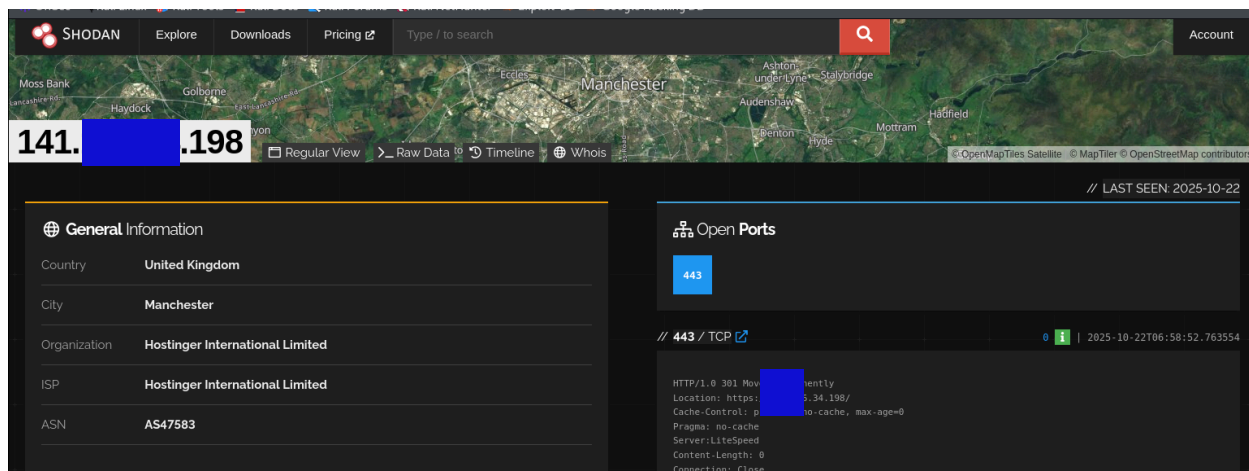
These findings essentially provide an attacker with a full roadmap for an attack; the OSINT report confirms that the government's digital perimeter is fully mapped and contains numerous points of entry that can be systematically tested and exploited.

Google Advanced Search Operators: which is also commonly referred to as Google Dorking (or Google Hacking). The search results indicate hits for documents related to the "Everything Technology" agenda in Anambra State.



The information displayed through (**Shodan.io**) search tool is a profile for a specific IP address (partially obscured) and provides details about its location, ownership, and what services are running on it.

Shodan.io



The screenshot shows the Shodan.io search results for the IP address 141.198.198.198. The interface includes a map of Manchester, UK, and a table of general information. The general information table lists the following details:

General Information	
Country	United Kingdom
City	Manchester
Organization	Hostinger International Limited
ISP	Hostinger International Limited
ASN	AS47583

Additional details shown include the IP address 141.198.198.198, the port 443, and the last seen date of 2025-10-22. The raw data section shows the following HTTP response:

```
HTTP/1.0 301 Moved temporarily
Location: https://141.198.198.198/
Cache-Control: public, no-cache, max-age=0
Pragma: no-cache
Server: LiteSpeed
Content-Length: 0
Connection: Close
```

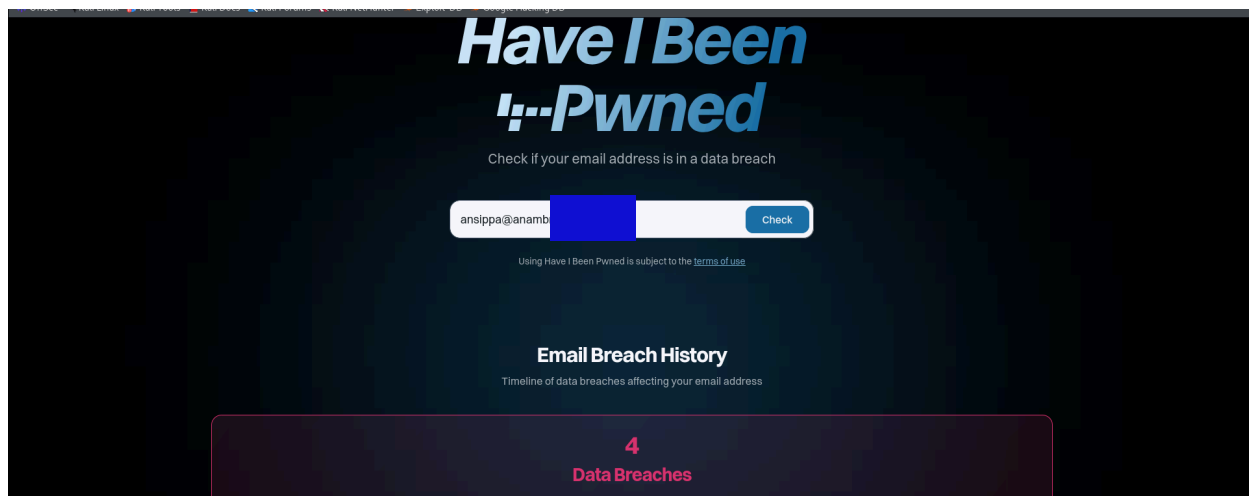
The result above confirms that the IP address belongs to a server hosted by Hostinger International Limited in Manchester, UK. It is running a LiteSpeed web server and is configured to handle secure web traffic (HTTPS), most likely acting as a redirect or the backend for a customer's website.

The image below displays the results of an email address search on the popular security website Have I Been Pwned? (HIBP).

This step is part of an ongoing OSINT (Open Source Intelligence) investigation.

The tool is used to check if an email address has been included in any publicly reported data breaches.

Haveibeenpwned.com



The screenshot shows the Have I Been Pwned (HIBP) website. The main heading is "Have I Been Pwned". Below the heading, it says "Check if your email address is in a data breach". A search bar contains the email address "ansippa@anamb" and a "Check" button. Below the search bar, it says "Using Have I Been Pwned is subject to the [terms of use](#)".

Below the search bar, there is a section titled "Email Breach History" with the subtitle "Timeline of data breaches affecting your email address". Below this section, there is a red box with the number "4" and the text "Data Breaches".

This result in the image above validates the email address found in the first OSINT step and confirms that the individual associated with it has had their digital credentials compromised multiple times. This makes them a high-priority target for a focused attack.

Totalvirus.com

The screenshot shows the Totalvirus.com interface. At the top, there's a search bar with '102.164.32.0/21' entered. Below the search bar, a green circle with '0' and '95' indicates a 'Community Score'. To the right, a message states 'No security vendor flagged this IP address as malicious'. Below this, the IP address '102.164.32.0/21' is listed with 'AS 37125 (Layer4)' and a 'NG' status. A 'Last Analysis Date' of 'a moment ago' is also shown. The interface has tabs for 'DETECTION', 'DETAILS', 'RELATIONS', and 'COMMUNITY'. Under 'DETAILS', there's a section for 'Basic Properties' listing network information: Network (102.164.32.0/21), Autonomous System Number (37125), Autonomous System Label (Layer4), Regional Internet Registry (AFRINIC), Country (NG), and Continent (AF). Below this, there's a section for 'Last HTTPS Certificate' showing the 'JARM Fingerprint' and 'Last HTTPS Certificate' details, including the signature algorithm (sha256RSA) and the signature itself.

The image above shows the result of checking one of the IP addresses discovered in the initial OSINT scan (from the first image) against a public intelligence platform, likely VirusTotal or a similar service that aggregates threat intelligence.

The image below lists various SSL/TLS certificates issued recently for the domain, revealing many active and historical subdomains. This is an extremely valuable resource for an attacker performing reconnaissance

Crt.sh

Criteria Type: Identity Match: ILIKE Search: 'anambastate.gov.ng'									
Certificates	cert.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities	Issuer Name		
	22320303040	2025-11-16	2024-11-15	2025-11-14	anagcc.anambastate.gov.ng	anagcc.anambastate.gov.ng	C=US, O=DigiCert Inc, CN=DigiCert ECC CA 2018		
	22320530295	2025-11-16	2025-11-16	2026-12-16	anagcc.anambastate.gov.ng	anagcc.anambastate.gov.ng	C=AE, O=SSL2BUY EMI, CN=SSL2BUY EMI		
	22320514726	2025-11-16	2025-11-16	2026-12-16	anagcc.anambastate.gov.ng	anagcc.anambastate.gov.ng	C=AE, O=SSL2BUY EMI, CN=SSL2BUY EMI		
	22446292747	2025-11-13	2025-11-13	2026-02-11	adminlens.anambastate.gov.ng	adminlens.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22446291620	2025-11-13	2025-11-13	2026-02-11	adminlens.anambastate.gov.ng	adminlens.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22445973810	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22446038105	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22443992203	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22442706091	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22443709827	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22442992173	2025-11-13	2025-11-13	2026-02-11	lens2.anambastate.gov.ng	lens2.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22439819754	2025-11-13	2025-11-13	2026-02-11	lens1.anambastate.gov.ng	lens1.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22439800830	2025-11-13	2025-11-13	2026-02-11	lens1.anambastate.gov.ng	lens1.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22356057624	2025-11-10	2025-11-10	2026-02-08	emoji.anambastate.gov.ng	emoji.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22356054149	2025-11-10	2025-11-10	2026-02-08	emoji.anambastate.gov.ng	emoji.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22356036245	2025-11-10	2025-11-10	2026-02-08	api-emoji.anambastate.gov.ng	api-emoji.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22356030880	2025-11-10	2025-11-10	2026-02-08	api-emoji.anambastate.gov.ng	api-emoji.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22345157354	2025-11-09	2025-11-09	2026-02-07	portal.ashefame.com	portal.ashefame.com	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22345156790	2025-11-09	2025-11-09	2026-02-07	portal.ashefame.com	portal.ashefame.com	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	22309453064	2025-11-08	2025-11-08	2026-02-06	*.anambastate.gov.ng	*.anambastate.gov.ng	C=US, O=Google Trust, CN=Google Trust		
	22309446082	2025-11-08	2025-11-08	2026-02-06	*.anambastate.gov.ng	*.anambastate.gov.ng	C=US, O=Google Trust, CN=Google Trust		
	22184656073	2025-11-03	2025-09-10	2025-12-09	*.anambastate.gov.ng	*.anambastate.gov.ng	C=US, O=Google Trust, CN=Google Trust		
	22087013816	2025-10-29	2025-10-29	2026-01-27	mail.assec.anambastate.gov.ng	mail.assec.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		
	50084000931	2025-10-26	2025-10-26	2026-01-23	mail.assec.anambastate.gov.ng	mail.assec.anambastate.gov.ng	C=US, O=Let's Encrypt, CN=Let's Encrypt		

THREAT ACTOR PROFILING

The vulnerability assessment performed on the Anambra State Government infrastructure reveals a broad attack surface, which attracts two main categories of threat actors: **organized cybercrime (Ransomware-as-a-Service, RaaS)** and **state-sponsored espionage groups**.

Below is a profile of 5 active, top-tier ransomware groups and an analysis of the likely threat actors that would target a Nigerian state government entity based on the observed vulnerabilities.

1.

GROUP NAME	LockBit
COUNTRY/AFFILIATION	Russia-linked
SPONSOR/FUNDING	RaaS Affiliates (Profit-sharing)
MOTIVATION	Financial/Extortion
DESCRIPTION	They were the most dominant RaaS group, targeting every sector globally.
MODEL OF OPERATION (TTPs)	Known for speed, using intermittent encryption to evade detection, exploiting vulnerabilities in VPNs and firewalls, and using double extortion (encrypting data and stealing it) In May 2024, the LockBit gang claimed responsibility for an attack on Canadian retailer London Drugs, which closed all locations across Canada. LockBit then threatened to release data within 48 hours if a ransom of \$25 million was not paid. London Drugs stated that they were "unwilling and unable to pay ransom" to the parties involved. No customer or primary employee data was compromised. On May 23, 2024, the company confirmed that data had been leaked by Lockbit, and that affected employees were being offered identity theft protection services.
SOURCES	• Cybersecurity and Infrastructure Security Agency

	(CISA) ● Federal Bureau of Investigation (FBI) ● Mandiant (Google Cloud) ● CrowdStrike ● Recorded Future
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2.

GROUP NAME	RansomHub
COUNTRY/AFFILIATION	Successor to other RaaS groups
SPONSOR/FUNDING	RaaS Affiliates (Aggressive profit-sharing)
MOTIVATION	Financial/Extortion
DESCRIPTION	Emerged quickly after disruptions to other major groups. Mode of Operation:
MODEL OF OPERATION (TTPs)	Highly aggressive RaaS model, often targeting large organizations. Known for speed, high ransom demands, and targeting companies that have insurance or critical operations. RansomHub was responsible for several high-profile attacks, including a re-extortion attempt on Change Healthcare and an attack on the auction house Christie's. As of April 2025, RansomHub's infrastructure and leak site went offline, and the group is currently considered inactive, with many of its affiliates reportedly migrating to other RaaS operations, most notably DragonForce
SOURCES	● Check Point ● Mjolnir Security ● BlackFog ● CISA ● Picus Security ● Cyber Swachhta Kendra

3.

GROUP NAME	Akira
COUNTRY/AFFILIATION	Possible link to former Conti group
SPONSOR/FUNDING	RaaS Affiliates
MOTIVATION	Financial/Extortion
DESCRIPTION	A highly sophisticated and resurgent group.
MODEL OF OPERATION (TTPs)	Often exploits VPN vulnerabilities (like SonicWall or Cisco), uses stolen credentials for initial access, and is known for bypassing Multi-Factor Authentication (MFA) on local accounts. The Akira ransomware group is a highly active, financially-motivated threat actor operating under the Ransomware-as-a-Service (RaaS) model. Since emerging in early 2023, they have consistently evolved their tactics, techniques, and procedures (TTPs) to maximize impact, frequently targeting small-to-medium businesses (SMBs) and large organizations across critical sectors globally.
SOURCES	● CISA ● IBMX-Force ● Fortinet ● SentinelOne ● Sophos

4.

GROUP NAME	BlackSuit, Zeon
COUNTRY/AFFILIATION	Russia
SPONSOR/FUNDING	Unknown
MOTIVATION	Financial/Extortion
DESCRIPTION	Royal ransomware group is an experienced, financially motivated cybercrime entity that emerged in early 2022. The group is known for human-operated ransomware attacks. Unlike many groups that use a Ransomware-as-a-Service (RaaS) model with affiliates, the core Royal group operated privately with a small, vetted team, or by purchasing access from initial access brokers (IABs).
MODEL OF OPERATION (TTPs)	Royal Mail (January 2023): Royal Mail, the GB's primary postal service, suffered a significant ransomware attack that caused a temporary suspension of its international delivery operations. The attack caused substantial financial losses for the company, which publicly declined to pay the attackers' ransom demand of approximately £65.7 million (around \$79.85 million USD at the time). City of Oregon (May 2023): The city of Oregon experienced a ransomware attack where county data was encrypted. The attack impacted all government operations except for election and emergency services dispatch systems. The cost of restoring the systems was estimated at millions of dollars, and the specific ransom amount demanded by Royal was not disclosed by officials.
SOURCES	• CISA • FBI • GLocal Law Enforcement • Threat Intelligence Firms

5.

GROUP NAME	Black Basta
COUNTRY/AFFILIATION	Russia-linked
SPONSOR/FUNDING	RaaS Affiliates
MOTIVATION	Financial/Extortion
DESCRIPTION	Active since 2022, quickly becoming a top-tier threat.
MODEL OF OPERATION (TTPs)	Attacks often begin with phishing or exploiting known vulnerabilities. They use sophisticated data exfiltration methods, prioritizing and categorizing sensitive information before the encryption stage. The Black Basta ransomware group has a global reach, but its operations are primarily concentrated on high-value targets in North America, Europe, and Australia. Since its emergence in April 2022, Black Basta has impacted over 500 organizations worldwide, successfully extorting tens of millions of dollars.
SOURCES	● CISA/FBI: Joint Cybersecurity Advisory ● AA24-131A (#StopRansomware: Black Basta),

TOOLS/TACTICS AND TECHNIQUES PROCEDURES USED BY “AKIRA” (MITRE ATTACK)

[Akira](#) is a ransomware variant and ransomware deployment entity active since at least March 2023. Akira uses compromised credentials to access single-factor external access mechanisms such as VPNs for initial access, then various publicly-available tools and techniques for lateral movement. Akira operations are associated with "double extortion" ransomware activity, where data is exfiltrated from victim environments prior to encryption, with threats to publish files if a ransom is not paid. Technical analysis of Akira ransomware indicates variants capable of targeting Windows or VMWare ESXi hypervisors and multiple overlaps with [Conti](#) ransomware.

TACTIC TECHNIQUE AND PROCEDURE

Domain	ID	Name	Use
Enterprise	T1531	Account Access Removal	Akira deletes administrator accounts in victim networks prior to encryption
Enterprise	T1560	Archive Collected Data: Archive via Utility	Akira uses utilities such as WinRAR to archive data prior to exfiltration
Enterprise	T1059	Command and Scripting Interpreter: PowerShell	Akira has used PowerShell scripts for credential harvesting and privilege escalation
Enterprise	T1486	Data Encrypted for Impact	Akira encrypts files in victim environments as part of ransomware operations
Enterprise	T1213	Data from Information	Akira has accessed and

		Repositories: Sharepoint	downloaded information stored in SharePoint instances as part of data gathering and exfiltration activity
Enterprise	T1482	Domain Trust Discovery	Akira uses the built-in Nltest utility or tools such as AdFind to enumerate Active Directory trusts in victim environments

AKIRA

RANSOMEWARE TOOLSET

I. Initial Access and Privilege Escalation

These tools are used to acquire high-level credentials and bypass security controls.

TOOL CATEGORY	SPECIFIC TOOL USED
Credential Harvesting	Mimikatz, Lazagne
Password Spraying	SharpDomainSpray
Exploitation	Veeam.Backup.MountService.exe

II. Reconnaissance and Lateral Movement

These tools help map the internal network and move from the initial compromised host to critical systems like domain controllers.

Tool Category	Specific Tools Used	Purpose
Network Scanning	SoftPerfect Network Scanner (<i>netscan.exe</i>), Advanced IP Scanner	Used to discover active network devices, shared folders, and network topology for planning the attack route.
Remote Access/Persistence	AnyDesk, LogMeIn, RustDesk, MobaXterm	Legitimate remote access software used to maintain a persistent presence, blend in with legitimate administrator activity, and control compromised systems.
Protocol Abuse	RDP, SSH, Impacket (<i>wmiexec.py</i>)	Used to pivot laterally across the network by abusing standard remote services and the Windows Management Instrumentation (WMI) protocol.

III. Defense Evasion and Command & Control (C2)

These tools are crucial for disabling security products and creating a covert communication channel.

Tool Category	Specific Tools Used	Purpose
Security Evasion	PowerTool, Native Windows Utilities	Used to terminate antivirus and EDR-related processes, uninstall EDR systems, and modify firewall settings to avoid detection.
Tunneling/C2	Ngrok, Cloudflare Tunnel	Used as reverse proxy tools to create encrypted tunnels, allowing the threat actors to maintain command-and-control communication while bypassing perimeter firewalls.

IV. Data Exfiltration and Encryption

These tools prepare for and execute the double extortion attack.

Tool Category	Specific Tools Used	Purpose		**Data Exfiltration**	**RCIone, FileZilla, WinSCP**	Used to transfer massive volumes of stolen data quickly to cloud storage services (like Mega) or external servers before the encryption phase.
Archiving	**WinRAR**	Used to compress stolen data prior to exfiltration, making the transfer faster and more efficient.				
Encryption	**Akira Ransomware Binary** (C++ and **Megazord/Rust** variants)	The custom payload used to encrypt files. It uses native commands (e.g., PowerShell) to **delete Volume Shadow Copies** to prevent system recovery.				
The consistent use of this toolkit, particularly the combination of legitimate remote access software (AnyDesk) with potent credential theft tools (Mimikatz), highlights Akira's operational maturity and focus on efficiency.

A consolidated summary of the General Findings from the Open Source Intelligence (OSINT) investigation provided, along with the Prioritized Mitigation Measures required to address the identified vulnerabilities.

The investigation revealed that the organization has a severely exposed digital perimeter, which substantially elevates the risk of a successful, high-impact ransomware attack (e.g., from groups like Akira or Black Basta).

GENERAL FINDINGS FROM “OSINT” INVESTIGATION

The analysis of the uploaded images revealed a wealth of public information that hostile actors, particularly financially motivated RaaS affiliates, could use to compromise the network.

A. Critical Attack Surface Exposure (176 Hosts)

- Finding: The initial scan and Certificate Transparency logs (crt.sh) confirmed the existence of 176 different subdomains/hosts (e.g., *adminens.*, *telemed.*, *portal.ashefarms.**).
- Risk: This means the organization's security depends on the strongest host out of 176, but the attacker only needs to find the weakest host. The administrative names (*adminens*, *adminsns*) indicate high-value, priority targets.
- Technology & Location: Specific IP addresses were identified, confirming the use of LiteSpeed servers and the location/ownership of network ranges (e.g., Hostinger, AFRINIC/Nigeria).

B. Confirmed Credential and Human Risk

- Finding: A list of official staff emails was compiled. The Have I Been Pwned? check confirmed at least one key government email (*ansippa@anar[REDACTED]*) was involved in four separate data breaches.

- Risk: The passwords for these accounts are highly likely to be available on the dark web. This enables attackers to execute credential stuffing against the 176 exposed hosts or craft highly convincing spear phishing attacks against the staff email list.

C. Technology Mapping and Vulnerability Indicators

- Finding: The investigation successfully mapped the infrastructure, identifying the use of specific web servers (LiteSpeed) and the ownership of IP ranges.
- Risk: This detailed mapping allows threat actors to select and deploy tailored exploits for the known software, accelerating the reconnaissance and initial access phases of an attack.

Prioritized Mitigation Measures

The recommended mitigation strategies are structured into three phases, prioritizing actions that block the most likely attack vectors (breached credentials and exposed hosts).

The goal is to stop the most immediate threat: using compromised credentials to gain access to the exposed infrastructure and leveraging known software vulnerabilities.

Phase 1. Block Compromised Credentials (CRITICAL)

ACTION	MITIGATION GOAL	PRIORITY
(a) Immediately enforce a mandatory password reset for all staff and system accounts, especially those identified on the staff email list.	Ensure the new passwords meet modern complexity standards and are unique.	High
(b) Immediately deploy or strengthen phishing-resistant Multi-Factor Authentication (MFA) across all exposed hosts, particularly the administrative ones (admins, etc.).	Use app-based MFA or hardware tokens. This renders any existing compromised passwords useless for initial access.	High
(c) Implement Account Lockout Policies on all exposed hosts to prevent automated credential stuffing attacks.	Set an aggressive threshold (e.g., 3-5 failed attempts) followed by a temporary lockout.	High

Phase 2. Harden and Patch Exposed Technology (CRITICAL)

ACTION	MITIGATION GOAL	PRIORITY
(a) Immediately verify that all identified LiteSpeed web servers and associated software (e.g., WordPress, PHP, database engines) are running the latest security patches.	Check for any Known Exploited Vulnerabilities (KEVs) specific to LiteSpeed or its common configurations and patch them first.	High
(b) Implement a Web Application Firewall (WAF) in front of the 176 exposed hosts.	A WAF can detect and block exploit attempts that target known vulnerabilities in the LiteSpeed or application layer, preventing the "tailored exploits" mentioned in the report.	Medium

Phase 3. Isolate and Restrict Access (CRITICAL)

ACTION	MITIGATION GOAL	PRIORITY
(a) For high-value administrative hosts, immediately implement IP whitelisting on firewalls/access control lists.	Only allow connections from known, trusted IP ranges (e.g., office VPN, security team hosts). This severely limits the attack surface for the administrative subdomains. These three steps directly block the most likely and highest-impact attack vectors: breached credentials and easily exploited, known software.	Medium

CONCLUSION AND STRATEGIC RECOMMENDATION

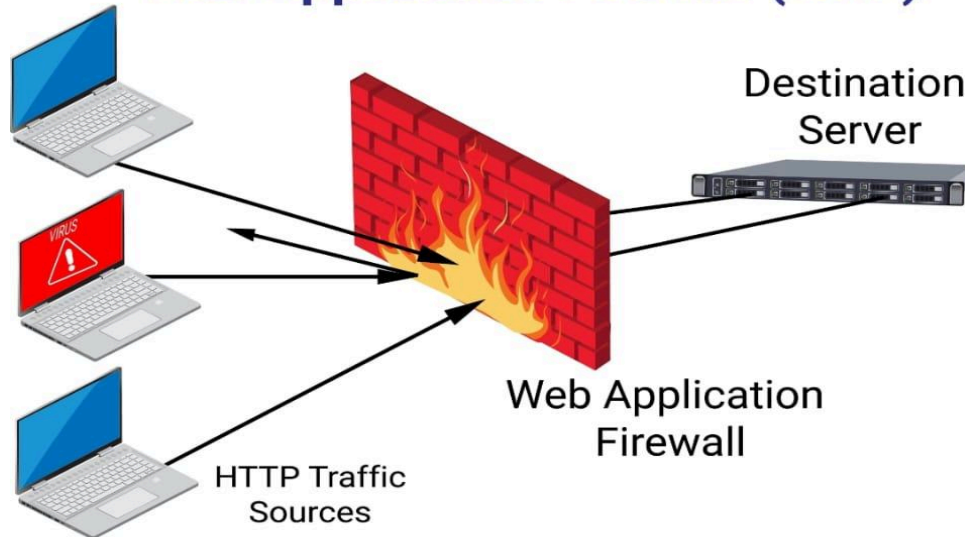
The threat intelligence and security risk assessment targeting the Anambra State Government, Nigeria, reveal a high-risk security posture driven by a confluence of common global cyber threats and specific regional vulnerabilities. The most critical finding is the demonstrated ease of reconnaissance and initial access for financially and politically motivated threat actors.

APPENDIX

This appendix provides a consolidated view of the Prioritized Mitigation Measures. The current security environment is marked by high exposure, low resilience, and critical dependencies on a large, weakly defended perimeter. The priority is not to stop every attack, but to block the most common and high-impact initial access vectors.

In summary, the organization cannot defend 176 front doors effectively. The immediate, critical requirement is to change the locks (MFA/Password Reset) and shut/barricade 90% of the doors (Attack Surface Reduction/IP Whitelisting). across all three phases, structured to directly block the confirmed attack vectors identified in the OSINT investigation and security risk assessment targeting the Anambra State Government.

Web Application Firewall (WAF)



References

Threat Intelligence and Specific Threat Actor Profiles
International Counter Ransomware Initiative. Available at:
<https://counter-ransomware.org/>

Have I Been Pwned?. Available at: haveibeenpwned.com
(Used to confirm credential exposure risk).

OWASP Foundation. OWASP Top 10 Web Application Security Risks (Used for hardening strategies related to the 176 exposed hosts and LiteSpeed technology).
Nigerian Information Technology Development Agency (NITDA). Guidelines and best practices for cyber security in Nigerian Government Ministries, Departments, and Agencies (MDAs)
<https://nitda.gov.ng/>

Nigerian Information Technology Development Agency (NITDA). Guidelines and best practices for cyber security in Nigerian Government Ministries, Departments, and Agencies (MDAs).
<https://nitda.gov.ng/>

Heimdal Security (2021). Conti Ransomware Leaks Police Citations and Forces Tulsa to Issue Breach Warning. Available at:
<https://heimdalsecurity.com/blog/conti-ransomware-leaks-police-citations-and-forces-the-city-of-tulsa-to-issue-a-data-breach-warning/>
(Accessed: 11 November 2025).

Purplesec (2022). Conti Ransomware Attack – Breach Report. Available at:
<https://purplesec.us/breach-report/conti-ransomware-attack/>
(Accessed: 12 November 2025).

International Counter Ransomware Initiative. Available at:
<https://counter-ransomware.org/>

Cybersecurity and Infrastructure Security Agency (CISA). Guidelines for Multi-Factor Authentication (MFA) and known exploited vulnerabilities (KEVs).

<https://www.cisa.gov/topics/cybersecurity-best-practices/multifactor-authentication>

Direct Link to CISA's Guide on Requiring MFA:

<https://www.cisa.gov/audiences/small-and-medium-businesses/secure-your-business/require-multifactor-authentication>

MITRE Corporation (2025). MITRE ATT&CK Framework. Available at:

<https://attack.mitre.org/>

(Accessed: 15 November 2025).

Shodan (2025). Shodan Search Engine: Internet-Wide Device Discovery. Available at:

<https://www.shodan.io>

(Accessed: 12 November 2025).

crt.sh (2025). Certificate Transparency Log Search Tool. Available at:

<https://crt.sh/...?q=nhs.uk&dir=^&sort=4&group=no>

(Accessed: 13 November 2025).

Cybersecurity and Infrastructure Security Agency (CISA). Guidelines for Multi-Factor Authentication (MFA) and known exploited vulnerabilities (KEVs).

International Counter Ransomware Initiative. Available at:

<https://counter-ransomware.org/>